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Combatting groundwater depletion in North-Western India: Obstacles, consequences, and water-saving agronomic approaches

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Abstract
The paper synthesizes agronomic, technological, and policy options relevant to groundwater sustainability in north-western India, a region of international significance for food security. Since the advent of the Green Revolution, there has been a significant shift from traditional crops to high-yielding, water-intensive crop varieties that require substantially more water. This transition has led to a dramatic increase in the number of tubewells and the overexploitation of groundwater resources. In the northwestern states of India, namely Punjab, Rajasthan and Haryana, the rate of groundwater extraction far exceeds natural recharge rates. This issue is aggravated by subsidized or free electricity for agriculture, which encourages excessive groundwater extraction. This paper reviews various agronomic and technological interventions and finds that measures such as laser land leveling, zero tillage, the System of Rice Intensification (SRI), direct-seeded rice (DSR), micro-irrigation, and water harvesting can reduce water use while maintaining yields. However, their effectiveness for long-term groundwater sustainability remains mixed. Among these, water harvesting structures appear most effective, offering dual benefits by reducing groundwater extraction and enhancing recharge. Furthermore, government policies, institutional support, and agricultural extension services play a crucial role in promoting the adoption of water-efficient practices. Also, international development cooperation can reinforce these efforts through concessional finance, technical assistance, and data sharing that lower adoption costs and strengthen monitoring. Effective implementation of these measures can help mitigate unsustainable exploitation of groundwater resources in Indian agriculture.

Keywords: Resource conservation technologies · Water balance · Water harvesting · Water smart technologies · Water use efficiency

Extended author information available on the last page of the article

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1 Introduction

Water, often called "the universal solvent," is both the most essential and most abundant liquid on Earth. Although around 70% of the earth's surface is covered by water, only 0.5% of the total is usable by humans (Baker et al., 2016), as 97.5% of world's water is contained in oceans and the remaining 2.5% is freshwater. Of this freshwater, approximately 69% is locked in glaciers and permafrost conditions, 30% exists as groundwater, and only 0.3% found is in lakes and rivers (Engelman & Leroy, 1993). Groundwater provides around half of global drinking water and 43% of total consumptive use (International Groundwater Resource Assessment Centre.www.un-igrac.org (accessed 10 Apr, 2020) 2020). However, uncertainties in the supply and management of water resources have increased due to climate change (Siebert et al., 2010). Significant global climate changes have resulted from human activities that have altered the composition of the atmosphere over the past few decades (IPCC, 2007; Malhi et al., 2021).

Groundwater resources account for approximately 85% of India's rural water demand, 50% of urban water demand, and over 50% of irrigation supply (National compilation on dynamic groundwater resources of India, 2023). Around 1000 km^3 of groundwater is extracted in the entire world out of which 67% is used for irrigation, 22% for domestic purpose and 11% for industry (International Groundwater Resource Assessment Centre.www.un-igrac.org (accessed 10 Apr, 2020) 2020). India extracts around 25% of global extracted groundwater and ranks first in groundwater extraction globally and uses 89% of it for irrigation (National compilation on groundwater resources of India, 2017). Since 1950, there has been a considerable increase in the rate of groundwater depletion (Konikov, 2011). Groundwater extraction in India witnessed substantial growth between 1960 and 1990, driven by the rapidly increasing population and the corresponding rise in water demand (Margat & Gun, 2013; Turner et al., 2019). The period of 2000–2008 showed an average of 145 $Km^3\ year^{-1}$ of extraction. A century-long assessment (1900–2008) estimated global groundwater depletion at approximately 4,534 km^3, of which about 1,361 km^3 occurred in northern India and its adjoining regions—accounting for nearly 30% of the total global depletion during that period (Konikov, 2011). A subsequent decadal analysis (2000–2012) revealed a continued decline, with groundwater levels falling across nearly 30% of the aquifers and an overall depletion estimated at 8.0 ± 3.0 km^3 in Indo-Gangetic basin. Furthermore, about 60% of the groundwater aquifers were found to be of poor quality due to high salinity and arsenic contamination (Macdonald et al., 2016).

India holds 4% of global water resources and receives an average annual rainfall of 1200 mm, yielding 4000 bcm of water from precipitation, with 690 bcm considered available surface water. Rainfall varies from 1100 to 90 mm, causing floods in some areas and droughts in others (Jain et al., 2020; Kumar et al., 2005; Ministry of water resources, 2017). India ranks 132nd globally in terms of per capita water availability, primarily due to its large population (Ministry of water resources, 2020). Per capita water availability has declined from 5200 m^3 in 1951 to 1545 m^3 in 2011, and is projected to further decrease to 1367 m^3 by 2031. Projected water demand in 2025 and 2050 is 1093 bcm and 1447 bcm, respectively, with irrigation comprising 83.30% and 74.08% of the total (India, 2018; Ministry of water resources, 2020).

India has total geographic area of 328.73 million ha out of which the net cultivated area and gross cultivated area is 140.13 million ha and 198.36 million ha, respectively while the net irrigated and gross irrigated area is 68.68 a and 96.46 million ha, respectively. Tubewells irrigation covers maximum area of 31.13 million ha (mha) followed by canals (16.28 mha), wells (11.31 mha), tanks (1.84 mha) and other sources (7.54 mha) in 2014–15 (Agricultural Statistics at a glance, 2022). The share of groundwater in irrigation exceeds 60% and has been uninterruptedly increasing over the years, driven by population pressure, high population density and cultivation of water-intensive and high value crops. The expansion of groundwater-based irrigation was examined using data from 16 major agrarian states in India covering the period 1951–2001 (Bhaduri et al., 2012).

This review paper examines the degree and patterns of groundwater depletion in India and analyzes how agricultural practices and government policies contribute to or mitigate this crisis. Drawing on scholarly research and government reports, the paper identifies and evaluates a range of possible responses to groundwater depletion. The study further assesses the feasibility of these approaches in northwestern India, exploring whether they function as complementary strategies or substitutes for one another. By synthesizing evidence from multiple sources, the review contributes to a clearer understanding of how sustainable agricultural practices and policy interventions can jointly support groundwater conservation and long-term agricultural sustainability.

The paper proceeds as follows: Sect. 2 outlines the methodology and analytical framework adopted for conducting the review. Section 3 provides an overview of groundwater recharge and extraction patterns in India. Section 4 examines the implications of groundwater depletion for the farm economy. Section 5 evaluates agronomic practices aimed at mitigating and adapting to groundwater scarcity. Finally, Sect. 6 concludes the paper and highlights the key policy suggestions derived from the analysis.

2 Materials and methods

PRISMA (Preferred Reporting Items for Systematic Meta-Analysis) framework was followed to conduct the systematic review (Page et al., 2021). Relevant studies were identified across electronic databases, including Google Scholar, Web of Science, and Scopus with searches limited to English-language publications. Groundwater-related studies were retrieved using keywords such as “groundwater extraction,” “water-saving technologies,” “groundwater depletion,” “groundwater recharge,” “crop-based water management strategies,” “economics of water-saving technologies,” “cost of groundwater extraction,” and “policies related to groundwater use.” Fig. 1.

For this review, study, the authors have selected three types of publications: (1) foundational studies, (2) recent research works (2005–2025), and (3) government reports. Relevant research papers and review papers were chosen based on publication in peer-reviewed, established journals (preference was given to high-impact Q1 and Q2 journals). Additionally, reports on groundwater use, rainfall, and groundwater extraction from Ministry of Water Resources, Ministry of Statistics and Programme Implementation, and Ministry of Agriculture & Farmers’ Welfare, were also incorporated into the study (Agricultural Statistics at a glance, 2022; Ministry of statistics & Programme implementation, 2023; Ministry

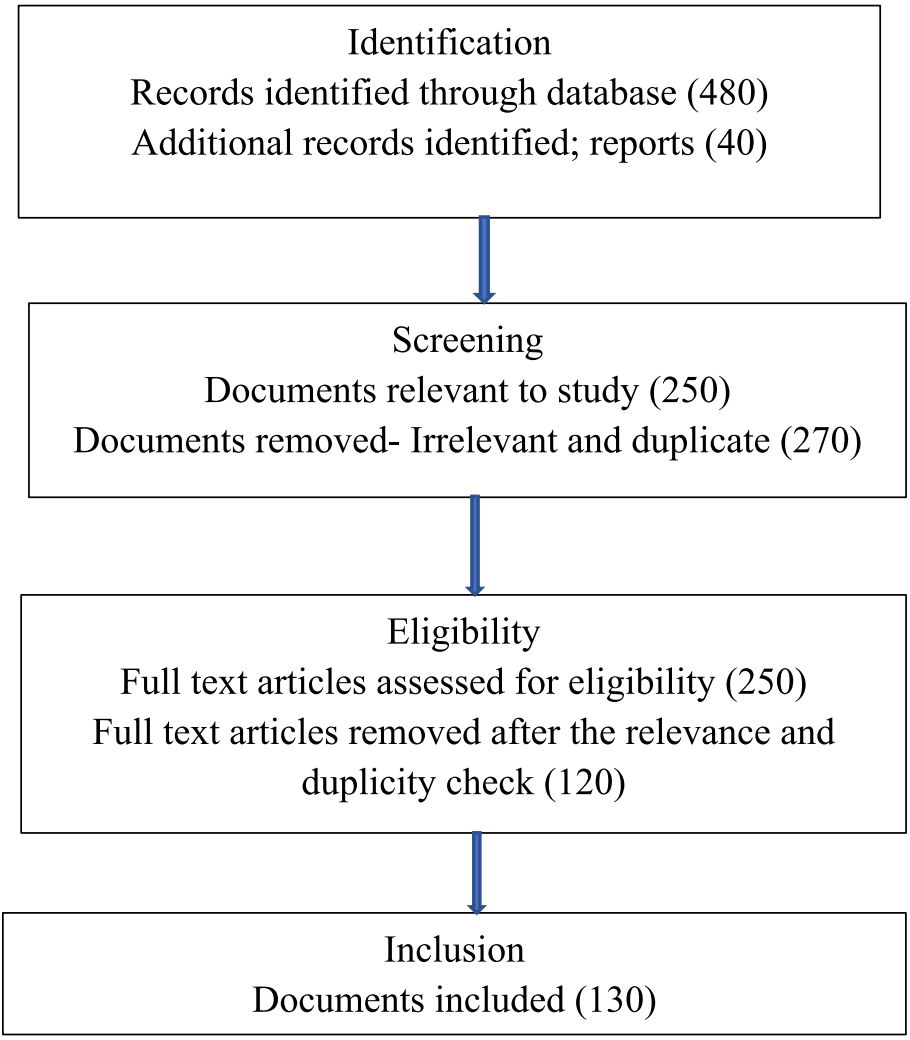


Fig. 1 PRISMA flow diagram of the study selection process

of water resources, 2017, 2020; National compilation on dynamic groundwater resources in India, 2011; National compilation on dynamic groundwater resources of India, 2020; National compilation on groundwater resources in India, 2022. 2022; National compilation on groundwater resources of India, 2017).

Beyond these bibliographic criteria, the authors rigorously assessed the methodological soundness of all selected studies. Studies that did not directly address the research context, or were found to be duplicate, overlapping, or irrelevant were excluded. All monetary figures originally denominated in Indian rupees have been converted to U.S. dollars using the 2022 exchange rate of ₹1 = $0.0127. Values originally reported in U.S. dollars are presented without conversion.

3 Groundwater recharge and extraction pattern in India

In 2022, the total annual groundwater recharge in India was 437.60 bcm, of which 398.08 bcm was extractable. Out of this, 239.16 bcm was extracted, resulting in groundwater stage of 60.08%. The largest share of recharge came from rainfall in monsoon season, contributing 241.35 bcm (55.15%), followed by recharge from other sources in non-monsoon season (89.07 bcm or 20.35%), recharge from other sources in monsoon season (82.30 bcm or 18.81%), and rainfall during non-monsoon season (24.88 bcm or 5.69%) (National compilation on groundwater resources in India, 2022. 2022). This highlights the critical role of monsoon rainfall in groundwater replenishment.

As shown in Table 1, rainfall patterns across India exhibit temporal variations that have been shifting over years due to climate change. Rising temperatures from global warming further increased evapotranspiration rates, thereby reducing the infiltration of natural precipitation (Shah, 2009). While greater precipitation generally enhances groundwater recharge, elevated evaporation rates and diminished infiltration linked to higher temperatures negatively affect groundwater replenishment.

Long-term variations in rainfall have altered groundwater storage over the past decade, both by directly affecting recharge and indirectly by influencing abstraction (Asoka et al., 2014). Since 2000, groundwater storage has declined rapidly, by ~ 70 km^3 in a decade, primarily owing to reduced rainfall and intensive well pumping; this is a sharp reversal of the 20th-century net gain of ~ 420 km^3 attributable to canal-induced recharge in north-western India and Pakistan (MacAllister et al., 2022). Even though the rainfall plays a vital role in groundwater recharge in India, there is spatial variation of groundwater recharge and it depends greatly upon the climatic situations, presence of water bodies, soil type, and aquifer system (National compilation on groundwater resources in India, 2022. 2022). The maximum annual groundwater recharge is observed in Uttar Pradesh due to the presence of Ganges river, soil type and rainfall received while the hill states of Jammu & Kashmir, Himachal Pradesh, Uttarakhand, and North eastern states except Assam have relatively low amount of groundwater recharge due to the hilly terrain.

In Punjab, Rajasthan and Haryana, annual groundwater utilization is more than the annual groundwater recharge causing negative water balance of −11.05, −4.39 and −2.08 bcm, respectively due to increasing urbanization, industrialization and concentration of acreage under water-guzzling crops (National compilation on groundwater resources in India, 2022. 2022). There are similar studies showing groundwater depletion rate of around 4.0 ± 1.0 cm $year^{-1}$ during 2000–2008 in Rajasthan, Haryana, Punjab and Delhi, which is mainly due to unsustainable water consumption for agricultural purposes, anthropogenic activities and climate change (Noori et al., 2023; Rodell et al., 2009).

Table 1 The temporal variation of rainfall in India (as observed in 2021–22)

Seasons	Months	Rainfall (in mm)
Winter	January, February	58.7
Pre Monsoon	March, April, May	130.6
South west Monsoon	June, July, August, September	874.6
Post Monsoon	October, November, December	177.8

Source: (Agricultural Statistics at a glance, 2022)

There is a slight reduction in groundwater extraction and consequent increase in water balance in different states of India over the years 2011–22. India has shown a decrease in groundwater extraction since 2011 by 3.10% and has resulted in 5.97% increase in water balance. However, Punjab, Haryana and Rajasthan are extracting more groundwater than their groundwater recharge and have reached the groundwater extraction stage of 165.99, 134.14 and 151.07% leading to negative water balance in these states. Although, Delhi and Punjab have shown more than 100% extraction stage during 2011, it has significantly declined till 2022 by 28.35 and 3.49%. Whereas, Rajasthan and Haryana have shown an increase of 10.27 and 0.86% groundwater extraction over 2011.

The Ministry of Water Resources has analyzed blocks all over India and classified them according to the quantum of water extraction and the extent of decline in water table over the years. The maximum proportion of overexploited blocks lies in the state of Punjab (80%) followed by Rajasthan (71%), Delhi (67%) and Haryana (61%) in 2011. The proportion of overexploited blocks has shown significant reduction in Delhi and Punjab which stands at 44.12% and 76.47% in 2022 while it increased slightly in Haryana (61.54%) and Rajasthan (72.52%). The classification of several blocks and their change over the years has been given in Fig. 2. There is a slight decline in the number of blocks deemed to be safe for groundwater extraction. However, the overexploited blocks were also decreased by 3%.

The analysis presents concerning statistics on groundwater depletion in India mainly northwestern states of India i.e., Punjab, Haryana, Rajasthan and Union territory of Delhi. The heavy dependence on groundwater for irrigation in India, especially for water-intensive crops, exacerbates the problem of groundwater depletion and contributes to unsustainable agricultural practices. Despite the recognition of groundwater depletion as a critical issue, there's a suboptimal use of effective means for sustainable water management, particularly in the agriculture sector.

4 Implication on farm economy

Groundwater depletion has far-reaching socio-economic implications. Irrigation accounts for about 70% of freshwater use in Asian countries. In India, irrigation represents roughly 80% of freshwater use, nearly half of which is used to irrigate rice (Singh et al., 2019).

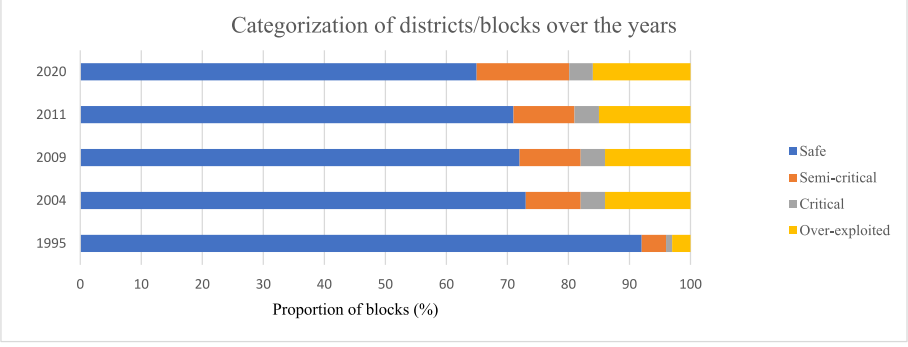


Fig. 2 Categorization of blocks over the years in India: Source: (Central Water Commission, 2022.www.cwc.gov.in accessed on 15 Jan 20242022)

Groundwater irrigation allows more precise control over the volume and timing of water application, typically leading to higher crop yields than surface water irrigation. In India, rice and wheat production could potentially increase by 36% and 3%, respectively, if all currently non-irrigated (rainfed) fields were irrigated with groundwater (Smilovic et al., 2015).

The irrigated area has expanded tremendously over the years, but this expansion has imposed an economic burden on farmers. The cost of groundwater extraction increases as the water table depth increases, and decreases as farm size increases (Srivastava et al., 2017). In Punjab, the government provides free electricity to farmers as an economic relief measure. However, this policy has had an unintended consequence: farmers have been over-irrigating their fields beyond necessary levels. The eventual removal of the electricity subsidy is expected to cause a 98% increase in the cost of groundwater extraction, thereby incentivizing farmers to adopt more efficient irrigation practices. Removing free electricity is estimated to result in the greatest reduction in groundwater use for maize (82%), followed by wheat (48%), sugarcane (41%), cotton (38%), and rice (29%) (Srivastava et al., 2017).

Moreover, a combination of factors – such as the prevalent paddy-wheat crop rotation, unreliable water supplies, and free electricity – has led to a long term groundwater level decline of about 41.6 cm per year (Baweja et al., 2017). In Haryana, electricity is supplied to farmers at highly subsidized rates, which has similarly led to excessive irrigation beyond agronomic needs. The power subsidy cost associated with surplus irrigation in rice cultivation was estimated at $158.26 per hectare. Additionally, the state government incurred a power subsidy of $0.11 billion for the extra irrigation of 7.23 million acre-feet, which accounted for 2.85% Haryana's revenue receipts and 2.31% of its development revenue expenditure in FY 2014–15 (Nehra, 2016).

A similar situation has been observed in southern India in context to groundwater exploitation. The cost of groundwater extraction has a significant effect on the farm income in Karnataka. The farm income is higher in villages categorized in safe blocks than villages of overexploited and critical blocks. However, the farmers of overexploited blocks showed highest technical efficiency in production of crops. The cost of groundwater extraction is maximum in overexploited villages followed by semi-critical villages and safe villages and it affects mainly small and marginal farmers due to less resources (Anantha, 2012). With the growing depths of the water table, the farmers of Tamilnadu have to incur additional costs for modification of wells which accounts for around 33–48% of actual borewell investment (Narayanamoorthy, 2015).

Excessive groundwater extraction by numerous farmers from a shared basin has led to significant groundwater decline, primarily due to the lack of widespread acceptance of conservation practices at the individual level. Groundwater decline may reduce cropping intensity by approximately 20% at the national level and by as much as 68% in regions experiencing severe groundwater depletion. (Jain et al., 2021).

Additionally, provision of subsidies has further worsened the issue (Ananda & Aheeyar, 2020; Strand, 2010). The flat tariffs and power subsidies provided by the government has raised the productivity of the crops at farm level but it has cause unnecessary substantial losses to the society as famers are investing heavily on tubewells depleting the groundwater. Moreover, even if the farms had to pay the marginal electricity cost, high net returns will persist in farming (Sayre & Taraz, 2018). A comparison of flat tariffs and metered tariffs in Punjab revealed that while flat tariffs are easier to administer, they do not incentivize farmers to use electricity judiciously. In contrast, metered tariffs may encourage efficient use

but are more complex and costly to implement (Sidhu et al., 2020). However, no empirical evidence was found to suggest that transitioning to meter-based billing would reduce water usage in Gujarat (Fishman et al., 2017). The shortcomings in tariff policies, combined with subsidies, have resulted in dual losses for Punjab; both through the decline of groundwater resources and imposition of substantial financial burdens on the state government (Sidhu et al., 2020).

The cultivation of crops not well-suited to the specific agro-climatic conditions, such as rice in Punjab and sugarcane in parts of Maharashtra, led to inefficient water use and contributes to excessive groundwater extraction. This practice is primarily driven by the availability of assured markets and lucrative prices for these crops. Consequently, the removal of power subsidies for agricultural use is unlikely to significantly reduce groundwater extraction (Dharmadhikary et al., 2018). The procurement of rice alone has contributed to at least 50% of the observed groundwater decline over a period of 34 years. Similarly, the wheat procurement in Madhya Pradesh has led to 5.3% increase in dry wells and 3.4% increase in deep tubewells showing the harmful externalities of well intentioned but poorly designed output subsidies (Chatterjee et al., 2024). However, the shift of crop procurement and production of crops of public distribution system (PDS) to other regions has the potential to arrest the groundwater depletion and even meeting the food targets and increasing the net farm income. This approach and associated reduction in electricity subsidies may balance regional farm incomes (Devineni et al., 2022).

The government of Gujarat has also started Jyotigram Yojna (JGY) scheme to provide rationed electricity supply for agricultural uses and non-rationed electricity supply to non-agriculture users to increase groundwater storage but the groundwater has continued to deplete in the area (Chindarkar & Grafton, 2018). However, there have found statistical evidence indicating that shift in the policies (reduction in electricity usage) in groundwater management in India has started refilling western and southern India's aquifers (Bhanja et al., 2017). Punjab government has also devised "The Punjab Preservation of Subsoil Water Act of 2009" to prohibit early rice transplanting so as to reduce unnecessary evaporation and consequently irrigation requirement. However, it was observed that the proportion of the state's area with groundwater levels deeper than 10 m increased from 30% in 2000 to 75% in 2019 (Sidhu et al., 2021). Haryana government has also enacted "Haryana Preservation of Sub-Soil Water Act, 2009" to reduce groundwater decline. But it hasn't led to desired results as evident through the water table decline and proportion of overexploited blocks over the years. The unavailability of related information and lack of monitoring of groundwater extraction and its availability is considered as major drawback for groundwater management (Shah, 2007). Groundwater overexploitation has socio-economic and ecological consequences like increased dependence of farmers, biasness or access inequity and serious health problems due to water contamination by fluorides (Kumar et al., 2001). However, groundwater is highly sensitive to changes in resource availability and associated costs, with rising extraction costs being most likely to reduce the rate of groundwater depletion in the twenty-first century [11, 133].

The cost of extraction of groundwater imposes a significant financial strain on farmers, predominantly small and marginal farmers, affecting their farm income and profitability. Subsidies provided by states, such as free electricity for farming, lead to enhanced groundwater extraction beyond sustainable levels, exacerbating groundwater depletion issues.

Individual-level conservation practices for groundwater are not common among farmers, further worsening the problem of groundwater depletion. Flaws in tariff policies and subsidies result in double losses, depleting groundwater resources and imposing financial burdens on governments. Rice and sugarcane cultivation in regions not suited for their growth leads to ineffective water use and exacerbates groundwater depletion. Despite the implementation of regulations like The Punjab Preservation of Subsoil Water Act of 2009, groundwater levels continue to decline, indicating ineffectiveness in addressing the issue.

5 Agronomic practices for mitigation and adaptation to groundwater depletion

This section evaluates agronomic practices for mitigating and adapting to groundwater scarcity. These include crop-based strategies, water-use planning and management, combined (layered) technologies, and data-driven solutions. Each is discussed below. In doing so, the study draws upon field experiments and long observational series, with particular attention to soils, climate, and farming systems, for performance that is often highly local. We also remark upon practical impediments, notably capital costs, informational shortcomings, and the sometimes perverse pull of prevailing policies and subsidies. Throughout, comparisons rest upon plain measures, such as irrigation requirement, water-use efficiency, energy per unit of yield, stability of output across seasons, and any discernible effects upon recharge and water quality. Finally, a modest word of caution is in order: improvements observed upon the individual holding do not invariably translate to aquifer-level relief, for reduced deep percolation and subsequent expansion of irrigated area may offset the gains; this caveat should be borne in mind as the paper proceeds.

5.1 Crop based mitigation strategies

5.1.1 Optimal cropping pattern

Given that more than half of India's gross cropped area is rainfed, the skewed allocation of irrigation water emerges as a primary cause of ineffective use of water resources. The depleting level of groundwater can be mitigated by avoiding the cultivation of water-intensive crops by changing the cropping pattern or crop diversification and improvement in crop cultivation to enhance water use efficiency. The rice crop, although has high productivity in regions of Punjab and Haryana, ranks very low in terms of water productivity. However, the eastern UP and Bihar has better irrigation water productivity which suggests the shifting of cultivation of rice from north-western to eastern states (Gulati & Mohan, 2018). The changing of cropping pattern to an optimal plan in Punjab has the potential to save 11.41% of water along with 0.64% increase in returns (Kaur, 2011). The strip intercropping of wheat/maize cultivated through conservation tillage with the use of straw mulching can raise the crop yields, raise water use efficiency in arid areas and reduce the carbon emissions (Hu et al., 2016). The in-situ moisture conservation practices such as fertigation, weed control, crop residue management etc. results in more productivity per drop of rain in pulses (Rajanna et al., 2017).

5.1.2 Resource conservation technologies (RCT)

Productivity of an area can be improved through resource conservation technologies chosen through multidisciplinary approach depending upon the region. In the central districts of Punjab, RCTs such as short duration varieties, delayed transplanting or sowing of crops, mechanical rice transplanting can improve the water and land productivity (Bhatt & Arora, 2015). The cultivation of medium duration rice varieties reduced the water use by 4–5% and improved the grain yield by 2–4% when sown directly than long-duration variety (Chaurasiya et al., 2022). The cultivation of short duration rice varieties reduces the evapotranspiration (ET) by 250 mm but also causes yield reduction of 2.5 t ha^{-1} in rice equivalent system compared to farmers practices (Singh et al., 2015). The yield of medium duration PAU-201 (photoperiod sensitive cultivar) is higher i.e., 7.8 t ha^{-1} over all the photoperiod insensitive cultivars whether medium duration PR-113 or short duration PR-115 and RH-257 by 14.1%, 12.8% and 11.5%, respectively. The cultivation of short duration varieties PR-115 and RH-257, however, results in 16.6% and 18.9% saving of water over PR-113. Delayed transplanting from 15 to 25 June or 5 July reduces the real water productivity (1.10–1.40 kg m^{-3}) and rice yield of all the photoperiod insensitive cultivars. However, the real water productivity of photosensitive cultivars is raised (1.63 kg m^{-3}) and yield remains statistically similar (Mahajan et al., 2009). The short duration varieties and late transplanting are most competent water saving technologies in area having declining water tables in water stressed regions of Punjab (Saini & Bhatt, 2019).

5.1.3 Direct seeded rice (DSR)

DSR is also an appropriate alternate to reduce crop water requirement. The cultivation of DSR reduces the greenhouse gases emission and thereby helps in adaptation and mitigation of climate change. It enhanced the organic matter turnovers, nutrient relations and carbon sequestration (Joshi et al., 2013). DSR has the potential to save 30–40% water as it eliminates the need for field puddling of fields and nursery preparation (Singh et al., 2012). Aerobic rice or DSR performs better in terms of yield under lowland conditions and also has better drought performance in upland conditions due to desired root traits. It saves 73% and 56% irrigation water during land preparation and growth period, respectively (Parthasarathi et al., 2012). The adoption of dry direct seeded rice reduced the grain yield by 3–6% but its irrigation water requirement is reduced by 23% over puddled transplanted rice (PTR) (Chaurasiya et al., 2022). DSR and transplanting with alternate wetting and drying (TRAWD) yields higher water productivity but lower yield when compared to transplanting under continuous flooding (TRCF) (Jabran et al., 2015). Moreover, the decreased yield in fine grain rice cultivars is mainly because of high panicle sterility. Irrigation water productivity was found to be significantly correlated and influenced with agronomic management practices, viz. irrigation, fertilization, agricultural pesticides and agricultural film etc. (Li et al., 2016).

DSR has also proven successful in reduction of cost, labour and water usage in Karnal district of Haryana. A reduction of 12–17% and 5–9% water usage as compared to puddled rice is observed in in 1 st year and 2nd year, respectively (Kumar et al., 2015). In Punjab, DSR is able to reduce irrigation water usage by 18–20% and labour requirement by 14 man-days ha^{-1}. The yield is reduced by 2–5% in the beginning but it overcome within 1–2 years.

The succeeding wheat crop however has 5.0 per cent higher yield in cropping sequencing with DSR in comparison to the sequencing with PTR (Bhullar et al., 2018). There is saving of 9% water with 14.6% higher percolation in DSR and thereby, can potentially contribute to groundwater recharge when compared to PTR (Gulati et al., 2021). However, weed infestation (Seem & Kumar, 2021), soil borne pathogens, nematode infestation, lodging, poor establishment of the crop, nutrient disorders, disease such as blast and brown leaf spot constraints adoption of DSR by the masses (Kaur & Singh, 2017).

5.1.4 System of rice intensification (SRI)

SRI method of cultivation can also decrease the water use as continued flooding is not needed in the rice as alternate wetting and drying of the rice fields do the purpose. The productivity of 10.39 tonnes ha^{-1} have been reported in some countries like India, Sri Lanka, China, Bangladesh with 23% water saving, whereas, in Japan, it can save about 28–35% irrigation water (Latif & Abdullah, 2012). SRI method saves water coupled with raising yield and lowering greenhouse gases emission and reducing the effect of global warming (Uphoff et al., 2013). The farmers have reduced their irrigation water use (40%) and cost of cultivation (23%) by adopting SRI. It also increases productivity by 46 per cent and additional net returns of \$545.12 ha^{-1} (Narayanmoorthy & Jothi, 2019). The farmers are quite receptive in adoption of SRI as it can lessen their water requirement, labour cost and reduce the conflicts arising due to water scarcity in the areas (Satyanarayana et al., 2007). Water productivity is found to be highest within the SRI-cultivation followed by alternate wetting and drying (AWD) and aerobic rice cultivation (Geethalakshmi et al., 2011).

These crop-based mitigation strategies are mainly focused on the crops and they have shown increased water use efficiency and water productivity. They have their role in decreasing water use and consequently reduce the groundwater extraction. However, their role in groundwater recharge is yet to be ascertained mainly because the role of these crop-based mitigation strategies is to decrease the crop evapotranspiration and to maintain or raise the crop yield while reducing water application or requirement.

5.2 Water use planning and management strategies

It is meant to enhance the water productivity or increase the water use efficiency (WUE) which can either be achieved by reduction of water requirement (adaptation) or water application (mitigation). Water management strategies can be planned on field scale as well as watershed scale (Shah, 2019).

5.2.1 Laser land levelling

With the help of laser land leveling technology, the farmers are able to save 21% irrigation water, 31% energy and significantly raises the yield of rice, wheat and sugarcane by 6.6, 5.4 and 10.9%, respectively in western Uttar Pradesh. The water productivity in the crops has improved by 33% and the net income of rice, wheat and sugarcane has raised by 14, 13.5 and 23.8% respectively (Naresh et al., 2014). Precision land leveling enhances the crop yield by increasing area under cultivation (1.5%–6%) and raising WUE and lead to enhanced water productivity and nutrient use efficiency (Jat et al., 2006). Similarly, around

16% saving in irrigation water was observed in laser land leveled fields along with increasing yields of rice and pigeon pea in Central India (Chaudhuri et al., 2007).

LLL has the potential to reduce groundwater extraction by 24% along with saving fuel due to reduced groundwater pumping. Among the available policy options, a first-hour LLL discount combined with second-degree price discrimination is the most effective strategy for groundwater conservation (Aryal et al., 2015). Farmers generally receive additional benefits of \$143.5 ha^{-1} yr^{-1} by increased yield and reduced electricity use when it is estimated in equivalence to average subsidized tariff for agricultural use by shifting from TLL to LLL in rice–wheat system. However, an increase of \$194 ha^{-1} yr^{-1} of additional benefits is projected to be obtained by the farmer if electricity tariff equivalent to average cost of supply is taken. Shifting of TLL to LLL will reduce the electricity consumption by 754 kWh yr^{-1} in rice–wheat system in Punjab and Haryana. If half of the area of rice–wheat system in Punjab and Haryana is shifted from TLL to LLL, it would provide additional benefits of \$385 million yr^{-1} (Lybbert et al., 2018).

The estimate of saving 1.5 million ha-m irrigation water is made if LLL is extended to 2 mha area under rice–wheat system in south east Asia which has monetary value estimated to be \$1400 million. The adoption of LLL will also increase the crop yields amounting to \$500 million over a period of three years and will reduce the GHG emission by 500 million kg. LLL is also estimated to increase the cultivable area by 3–6% because of reduction of bunds and channels in the fields (Jat et al., 2006; Sadik-Zada & Jalabi, 2025).

5.2.2 Microirrigation system

The micro-irrigation system (MIS) has the potential of saving water along with increasing crop productivity and reducing cost of irrigation. For the popularization of MIS (drip and sprinkler irrigation) and spread of water efficient technologies, Indian government is promoting Per Drop More Crop (PDMC) component of Pradhan Mantri Krishi Sinchayi Yojana operationized from 2015–16 (Anonymous., 2020). However, the effect of MIS on the sustainable use of groundwater varies according to the productivity gain by shifting to MIS from conventional methods, usage of saved water and the number of farmers adopting MIS (Namara et al., 2007). Micro-irrigation systems (MIS) currently irrigate approximately 10% of the potential area (Suresh et al., 2019), despite being recognized as an efficient technology for water conservation, increasing crop yields, improving WUE, reducing fertilizer and manure costs, and minimizing weed infestation (Priyan & Panchal, 2017). However, the adoption and use of MIS are limited by several factors. Despite widespread installations, its usage remains significantly lower in certain districts of Gujarat during the summer, even with impressive adoption rates, primarily due to groundwater scarcity (Viswanathan & Bahinipati, 2015).

Water requirement of rice irrigated with drip irrigation ranges between 938 to 1836 L kg−1 of rice in comparison to 4250–5508 L kg^{-1} in transplanted rice under surface check basin irrigation (Singh et al., 2019). The adoption of drip and sprinkler irrigation has the potential to reduce 2/3rd of groundwater extraction however, nearly half of these savings are offset by the subsequent expansion of the irrigated area. (Fishman et al., 2015). However, adoption of efficient irrigation technologies can lower the aquifer recharge by decreasing return inflows and can actually worsen the condition (Ward & Velazquez, 2008). The drip irrigation has found to save 32 per cent irrigation water and 40 per cent electricity usage

in Brinjal crop in Tamilnadu. The fertilizer cost has been reduced by 31 per cent while the productivity has been raised by 52% thereby leading to an increase of 54% in net returns (Narayanmoorthy et al., 2018). The highest WUE is observed in rice and wheat under drip irrigation and drip + rainport treatment, respectively (Meena et al., 2015). The drip irrigation was promoted as a climate resilient technology so as to lower down the demand for groundwater by increasing its efficiency but it has been observed that expansion of drip irrigation in Rajasthan has increased the groundwater over-extraction (Birkenholtz, 2017). However, drip irrigation is likely to raise the irrigation efficiency (50–90%), energy efficiency (30.5%) and fertilizer use efficiency (28.5%) (Priyan & Panchal, 2017).

The drip irrigation system in Tamilnadu has found to significantly increase the cropping intensity and irrigation intensity. A significant change in the cropping system has been observed as annual crops have been exchanged by coconut, banana, grapes etc. Moreover, drip irrigation has significantly reduced the cultivation cost and raised crop productivity and consequently, farm profitability (Kumar & Palanisami, 2010). With the provision of subsidies, the initial adoption of MIS was displayed but adoption at aggregate level was still low in terms of acreage and number of farmers in Gujarat (Bahinipati & Viswan.athan, P. K., 2017). Despite providing subsidies to the farmers, the small farmers are facing high capital costs as the drip providers behaves as rent seeking agents and increase the prices by 40% or more (Malik et al., 2016).

5.2.3 Alternate wetting and drying (AWD)

Continuous flooding of rice fields is a primary contributor to the high water demand of the rice crop; therefore, AWD presents a viable alternative for reducing rice water consumption. The application of irrigation water under AWD decreases compared to daily irrigation in both DSR and PTR systems. It was mainly due to less seepage, runoff and deep drainage. Among both these establishment methods, the irrigation input is lesser in DSR (Yadav et al., 2011). AWD reduces the grain yield by 3–14% but increases the WUE and lowers the irrigation water use by 26–52% (Chaurasiya et al., 2022). Irrigation WUE is 59% higher in AWD when compared with conventional system along with maintained yield. AWD has also shown the potential in reducing withdrawal from Mississippi river valley alluvial aquifer (Atwill et al., 2020). It increases the water use efficiency by 56% in Arkansas with 45–90% reduction in global warming potential (Nalley et al., 2015). AWD leads to strengthened nitrogen transformation process and preferential flow leading to decreased efficiency in water saving and increased NO_3^- N loading to the groundwater (Tan et al., 2012). It also reduces the irrigation hours by 38% without significant decrease in yields and profits. This reduced irrigation period corresponds to saving irrigation water and pumping energy (Rejesus et al., 2011).

5.2.4 Innovative water management

The application of other innovative water management technique like canopy cooling applied during the period of evaporative cooling has resulted in lower leaf temperature and higher value of photosynthesis compared to conventional method. The combination of evaporative cooling and precision irrigation thus led to higher water productivity along with increased yield, higher earliness in globe artichoke (Deligios et al., 2019). A combination of

four water management techniques viz. runoff diversion, on-site rainwater harvesting and conservation tillage were assessed and these resulted in enhanced maize yield upto 4.8 t ha^{-1} i.e., 3.8 t ha^{-1} more than current average of 1 t ha^{-1}. The mean available water productivity ranges between 0.35 and 0.51 kg m^{-3} (Makurira et al., 2011).

5.2.5 Water harvesting

The shifting of farmers to irrigation with on-farm reservoirs instead of groundwater pumping can help in reducing groundwater exploitation, fuel combustion and greenhouse gases emission and help in building of a healthier ecosystem [102, 132]. The interaction of surface water irrigation (tank irrigation) and groundwater level is very evident and thus for both well irrigation and tank irrigation must be used as complements to main hydrological balance (Chowdhury & Behera, 2018). Even though there is technical competence to reverse the water table decline in India, the actual impact depends upon the farmers' decisions of water use (Fishman et al., 2015). Water harvesting also seems to be a viable option for groundwater recharge. The natural recharge due to rainfall was estimated between 11 to 16% of annual rainfall. The recharge in the area was observed 390.29 ha m of which 38.53% is contributed by water harvesting structures (in Jamka micro-shed) (Gontia & Patil, 2012). In the city of Chennai, a widespread installation of rainwater harvesting structures led to a clearly observed rise in urban groundwater levels, reversing a declining trend (Jebamalar et al., 2012).

Also, examples from abroad illustrate what is possible: in the North China Plain, a severe groundwater decline was halted and reversed by a combining large-scale surface water transfers from the humid south and strict groundwater pumping regulations, aided by a series of wetter years. Groundwater levels there have been increasing at an average rate of ~ 0.7 m $year^{-1}$ demonstrating that integrated water management and strong policy interventions can achieve aquifer recovery across extensive regions (Long et al., 2025).

5.2.6 Irrigation scheduling and deficit irrigation

Application of irrigation according to tensiometer is able to save 13.0% water and power usage, cuts variable cost by 7.0% without any reduction of yield in Punjab. The state will be able to save 0.67 mha-m of water and save US$ 459 million if 30% of paddy area of the state is irrigated according to tensiometer based scheduling (Vatta et al., 2018). The use of tensiometer based irrigation is able to save water by 11–31% and 52–127 Kwh $acre^{-1}$ of power without any adverse impact of the grains and thereby, improves the overall water productivity (Bhatt et al., 2016). Deficit irrigation is the need of agriculture in the conditions of water scarcity. Regulated deficit irrigation (RDI) can raise the water productivity along with increasing farmers' profits in water limited areas (Fereres & Soriano, 2006). The scheduling of irrigation is found to enhance the irrigation water use efficiency (IWUE) and thereby, increase the contribution of groundwater to water use of the crop (Gao et al., 2017). The Sunflower crop sustains even with 35% deficit irrigation by utilizing shallow groundwater with reduced yield of 20% mainly in arid regions (Eltarabily et al., 2020). The deficit irrigation didn't stimulate the expansion of root system in clay soil and thereby, reduced the transpiration rate below its maximum rate. Therefore, an appreciable amount of water uptake doesn't occur from groundwater as cotton root system is highly sensitive to high soil density (Cohen et al., 1995).

5.2.7 Moisture conservation practices

Ridge furrow mulching (RFM) and rotational tillage (RT) has raised the WUE of wheat by 20.11 and 12.50%, respectively causing increase in the yield of wheat by 18 and 15%, respectively in China (Wang & Shangguan, 2015). The mulching of whole plots has raised the maize grain and straw yield by 58.6 and 35.0%, respectively in north western India (Arora & Bhatt, 2006). The manual tillage has high water holding capacity but lower water use efficiency due to high evaporation whereas zero tillage has low application efficiency due to high runoff (Okunade, 2006). Zero tillage in northern India has been found efficient in reducing water input without reducing the wheat yield when sown directly in rice stubbles due to provision of mulch benefits like regulation of soil temperature, hindering of sunrays, reducing wind speed near soil surface, improving soil health etc. (Bhatt et al., 2017). There is a significant effect of tillage method and irrigation level on the wheat yields in Bangladesh whereas water use efficiency increased with decrease of water use in irrigation (Sarker et al., 2012).

Evidence suggests that water management technologies such as LLL, AWD, and MIS can reduce groundwater extraction and improve water use efficiency. However, MIS has been associated with reduced groundwater recharge, and its widespread adoption has led to an expansion of irrigated areas, resulting in increased groundwater withdrawal despite improved efficiency. Deficit irrigation enhances water use efficiency but may negatively impact crop growth, particularly in clay soils, where it fails to stimulate root development in crops like cotton. In contrast, water harvesting structures provide dual benefits by reducing groundwater extraction and contributing to groundwater recharge. Moisture conservation techniques have also shown higher water use efficiency. Overall, water management technologies demonstrate improved or comparable crop yields to conventional methods, with the exception of deficit irrigation in certain soil types. Among these, water harvesting structures stand out as an effective approach for balancing water use with groundwater sustainability.

5.3 Layering of water smart technologies

The layering of technologies, where, two or more complementary technologies are adopted, there is sufficient evidence of increased water use efficiency. The adoption of aerobic rice with MIS can be a sustainable alternative to other rice production methods in water scarce areas leading to higher benefits and environmental safety by reducing methane emission (Parthasarathi et al., 2012). Deficit irrigation is also considered to have higher water use efficiency if used with water conservation techniques. The yield of maize crop was reported 1.6 times higher in ridge and straw mulched furrows (RF) and straw mulch (SM) applied with deficit irrigation (Zhou et al., 2011).

Technologies such as LLL, AWD, delayed transplanting, raised beds, short duration rice varieties reduce the irrigation water use but it is more likely to reduce deep drainage with minute or no effect on evapotranspiration. Although, reduction of deep drainage has absolute impact on reducing energy consumption for groundwater pumping, leaching, nutrient losses and groundwater pollution (Humphreys et al., 2010). The layering of smart technologies such as dry direct seeded rice, cultivation of medium-duration hybrid rice variety and AWD improved the water productivity by 21% and reduced the irrigation water usage

by 18% without any significant loss in grain yield (Chaurasiya et al., 2022). In case of rice–wheat cropping system in Indo-Gangetic plains, raised beds are not always efficient in reducing the amount of irrigation water and their effect varies according to soil type and age of the bed, depth to water table, size and shape of fields etc. (Kukkal et al., 2010). However, direct seeded rice (DSR) when cultivated on raised beds has the potential to save 13–23% irrigation water and have higher water use efficiency of 0.45 g L^{-1} but it also led to yield loss of 14–25% (Bhushan et al., 2007). The water use efficiency of sugarcane has improved by changing the row spacing, irrigation scheduling, adding a layer of crop residue and cultivation of suitable varieties. The addition of a layer of crop residue increased the water use efficiency by 18% without negatively affecting the yield (Olivier & Singels, 2015). Promotion of low water use crops, micro-irrigation, a limit on number of irrigation wells, efficient water and crop budgeting, awareness programs, effective management services for irrigation can help in groundwater conservation in India (Chandrakanth et al., 2011).

Some components of water use planning and management strategies are contributing to groundwater recharge while others are increasing water use efficiency or productivity or both. The impact of these technologies on groundwater recharge or reversing the groundwater decline has not yet quantified and there is technical incompetence of implementation of some of the technologies in a number of scarce areas. However, there are empirical evidence of groundwater recharge caused by large scale surface water diversion and stringent regulations. These water saving technologies are quite promising and are gaining popularity. However, their adoption is not as expected because of various constraints at farmers' personal, economic and social level. These constraints need to be removed to escalate their adoption among masses. Farmers must also be made aware about influence of their activities on the environment and the projected positive implications of adoption of water saving technologies on both environment and their own economy.

5.4 Data driven solutions for optimized groundwater utilization

The statistical models aim at optimization of the use of available resources and ecological parameters for sustainable development of agriculture. They are developed for governing the groundwater flow in combination with optimization techniques like quadratic and linear programming. These models are either for water policy evaluation or water allocation (Gorelick, 1983). To optimize water utilization, a linear programming model was formulated, and the proposed model is capable of achieving a reduction in water use by 28.51% (Difallah et al., 2017). A link between individual domains model is devised to formulate a multidisciplinary integrated model through open modeling interface, which was very efficient in building agricultural-groundwater-economic integrated model which in turn shows the influence of water policy in agricultural systems. This model shows that impact of agricultural policies varies spatially throughout the study region (Bulatewicz et al., 2010).

GIS based modelling is used for evaluation for potential of groundwater conservation through alternative management systems in western Nile delta. Through reduction of surface water inflow and raising the annual groundwater abstraction by 450 million m^3 has the potential to increase the net aquifer recharge by 5.7% and decrease its potentiality by around 91%. While, the annual aquifer potentiality can be increased by around 23% with construction of a new canal (Dawoud et al., 2005). In southeastern Louisiana, an improved non-dominated sorting genetic algorithm (NSGA-II) was linked with a high-infidelity groundwater

model (MODFLOW-2005) to determine the optimal extent of groundwater extraction with a focus to fulfil the main objectives related water supply, environmental issues and energy cost and was proven quite appealing (Yin et al., 2020). Integrated AnnAGNPS-MODFLOW modelling approach also demonstrated that optimizing irrigation strategies can enhance aquifer sustainability without compromising productivity. The integrated framework serves as decision support tool for evaluation of long term impacts of agricultural practices on water resources and promoting sustainable water management (Momm et al., 2022). In Jhajjar, Haryana, SaltMod model was used to analyze the impact of different management strategies on water table depth and groundwater salinity. It showed that groundwater levels are rising by about 0.7 m per year under current conditions. The study recommended reducing rice cultivation, cutting down canal water use and augmenting groundwater utilization. The study also resulted in formulation of optimal scenario in which minor adjustments of 2–8% in various inputs could prevent further waterlogging and secondary salinization (Singh, 2024).

The statistical modelling for determination of optimal water application in water scarce areas or determining the impact of various policy alternatives is quite beneficial in reducing groundwater extraction. The statistical models usually draw inferences based on small representative samples. So, the technical adoption or competence of various alternatives need to be judged before its implementation and promotion.

6 Conclusion and policy implications

India is currently extracting about 60.08% of its annual renewable groundwater (as of 2022). In critical regions such as Punjab, Haryana, and Rajasthan, groundwater extraction exceeds annual recharge, leading to persistent depletion of aquifers. In 2022, roughly 76% of assessment blocks in Punjab, 72% in Rajasthan, 61% in Haryana, and 44% in Delhi were over-exploited. Groundwater depletion has serious implications for the farm economy: as the water table drops, the energy and capital cost of extraction rises, disproportionately affecting small and marginal farmers. Furthermore, subsidized electricity for agriculture, while providing short-term relief to farmers, has incentivized inefficient water use and the cultivation of water-intensive crops (such as rice in northwestern India) in unsuitable areas. This not only accelerates groundwater decline but also imposes a financial strain on state governments due to rising subsidy burdens.

A range of agronomic practices and technologies have the potential to reduce groundwater withdrawals by lowering crop water requirements, reducing water applications, or increasing water-use efficiency. Crop-based mitigation strategies (optimal cropping patterns, RCTs, SRI, DSR, etc.) generally improve water-use efficiency and can maintain or improve yields, but their impact on groundwater storage is indirect. On-farm adoption of practices like laser land leveling, zero tillage, SRI, DSR, sprinkler/drip irrigation, water harvesting, and other resource-conservation technologies in regions like Punjab and Haryana has been shown to reduce water use without sacrificing yields. However, some efficiency improvements (e.g., micro-irrigation) can inadvertently reduce groundwater recharge and may even lead to greater total water use if farmers expand their irrigated area. Similarly, deficit irrigation can reduce water use, but often at the cost of some yield reduction, especially in heavy soils.

The earlier regulatory measures aimed at controlling paddy cultivation, including the Punjab Preservation of Subsoil Water Act (2009) and the Haryana Preservation of Sub-Soil Water Act (2009), have shown limited effectiveness in mitigating groundwater depletion. Between 2011 and 2022, the share of overexploited blocks in Punjab fell by only 4%age points, and in Haryana there was little to no improvement in overall extraction rates. The limited impact of these regulations is largely due to farmers' continued preference for water-intensive paddy cultivation, driven by assured procurement and minimum support prices that make rice lucrative despite its high water demand. This indicates that economic incentives are as important as regulations: without viable remunerative alternatives, farmers have little motivation to shift away from rice.

Going forward, a multi-pronged approach is needed to address India's groundwater crisis. Lessons can be drawn from historical instances of aquifer recharge in India such as the recharge that occurred via canal irrigation in the mid-twentieth century and international examples like China's successful groundwater regulation and surface water import in the North China Plain. Key strategies should include: (a) promoting technology-driven interventions (e.g., modern irrigation methods, precision agriculture, water-saving crop varieties), (b) incentivizing crop diversification away from water-intensive crops in over-exploited areas, (c) implementing rational water pricing or electricity pricing to encourage conservation while protecting vulnerable farmers, and (d) strengthening community-based and regulatory mechanisms for groundwater management such as pumping quotas or collective management of aquifers.

By coupling these interventions with robust extension services and farmer education on the environmental and long-term economic benefits of water conservation, India can move toward a balanced use of groundwater. Calibrated reduction of energy subsidies, or their redesign into pro-conservation incentives, together with investment in recharge infrastructure and the development of markets and value chains for less water-intensive crops, can curb over-extraction. International development cooperation can amplify these efforts through concessional finance for recharge and efficiency projects, technical assistance to strengthen monitoring and regulation, shared use of satellite and modeling tools, and South–South knowledge exchange on crop diversification and demand management. Partnerships with multilateral banks and climate funds can support pilot programmes, results-based incentives, and blended-finance mechanisms that lower adoption costs for smallholders. Only through an integrated approach that combines on-farm practices, technological innovation, economic incentives, regulatory frameworks, and well-designed international support can sustainable groundwater management be achieved, safeguarding both agricultural livelihoods and the resource base for future generations.

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